

The slide features a light blue background with decorative dark blue geometric shapes in the corners. These shapes include squares and rectangles with diagonal lines, creating a modern, abstract design.

AI-DRIVEN HR INTELLIGENCE SYSTEM

Tokanaeva Aisana,
Zharova malika

BUSINESS PROBLEM



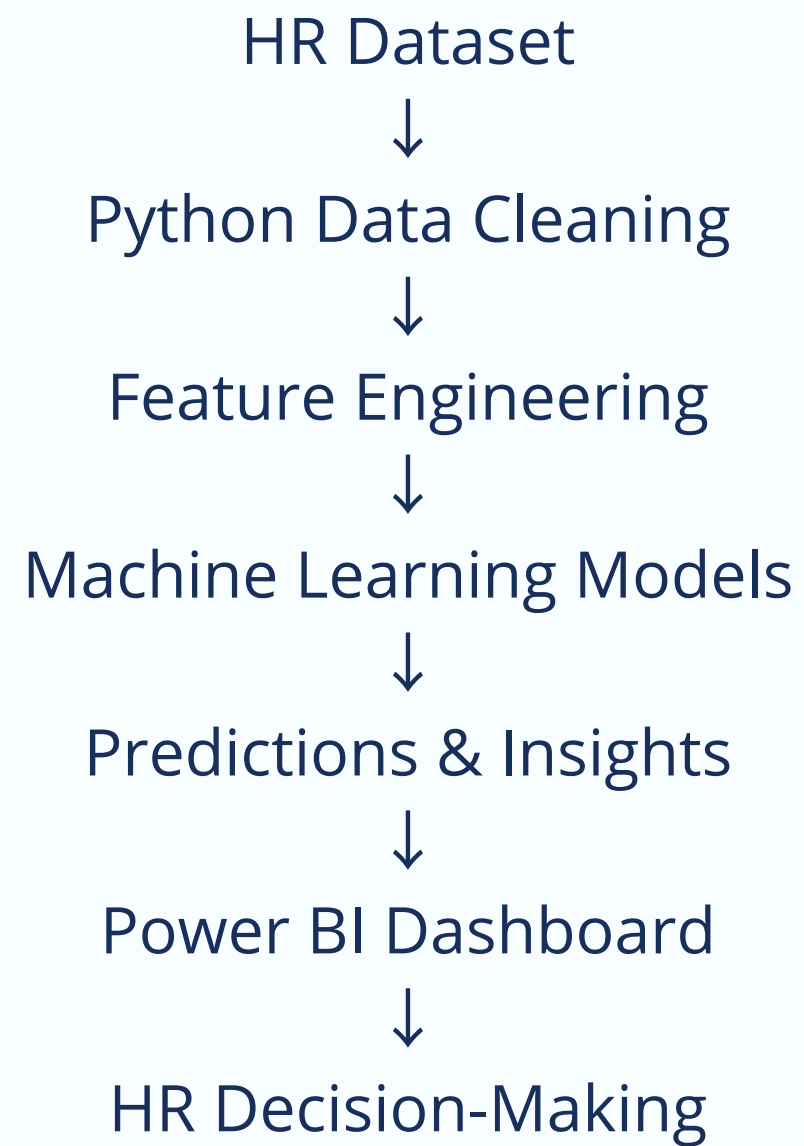
PROJECT OBJECTIVES

Project Objectives

- Clean and prepare HR workforce data
- Analyze workforce trends and risks
- Build AI hiring recommendation model
- Predict employee attrition
- Segment employees into performance groups
- Forecast hiring demand
- Audit fairness and GDPR compliance
- Build interactive Power BI dashboard



SYSTEM ARCHITECTURE



DATA PREPARATION

Data Preparation Steps

- ✓ Missing value handling
- ✓ Duplicate removal
- ✓ Category standardization
- ✓ Date conversion
- ✓ Feature engineering

Engineered Features

- YearsAtCompany
- Engagement_Risk_Score
- Salary_Vs_Dept_Avg
- Overtime_Risk

Dataset Shape:
(311, 36)

Dataset Preview:

	Employee_Name	EmpID	MarriedID	MaritalStatusID	GenderID	\
0	Adinolfi, Wilson K	10026	0	0	1	
1	Ait Sidi, Karthikeyan	10084	1	1	1	
2	Akinkuolie, Sarah	10196	1	1	0	
3	Alagbe,Trina	10088	1	1	0	
4	Anderson, Carol	10069	0	2	0	

	EmpStatusID	DeptID	PerfScoreID	FromDiversityJobFairID	Salary	...	\
0	1	5	4		0	62506	...
1	5	3	3		0	104437	...
2	5	5	3		0	64955	...
3	1	5	3		0	64991	...
4	5	5	3		0	50825	...

	ManagerName	ManagerID	RecruitmentSource	PerformanceScore	\
0	Michael Albert	22.0	LinkedIn	Exceeds	
1	Simon Roup	4.0	Indeed	Fully Meets	
2	Kissy Sullivan	20.0	LinkedIn	Fully Meets	
3	Elijah Gray	16.0	Indeed	Fully Meets	
4	Webster Butler	39.0	Google Search	Fully Meets	

	EngagementSurvey	EmpSatisfaction	SpecialProjectsCount	\
0	4.60		5	0
1	4.96		3	6
2	3.02		3	0
3	4.84		5	0
4	5.00		4	0

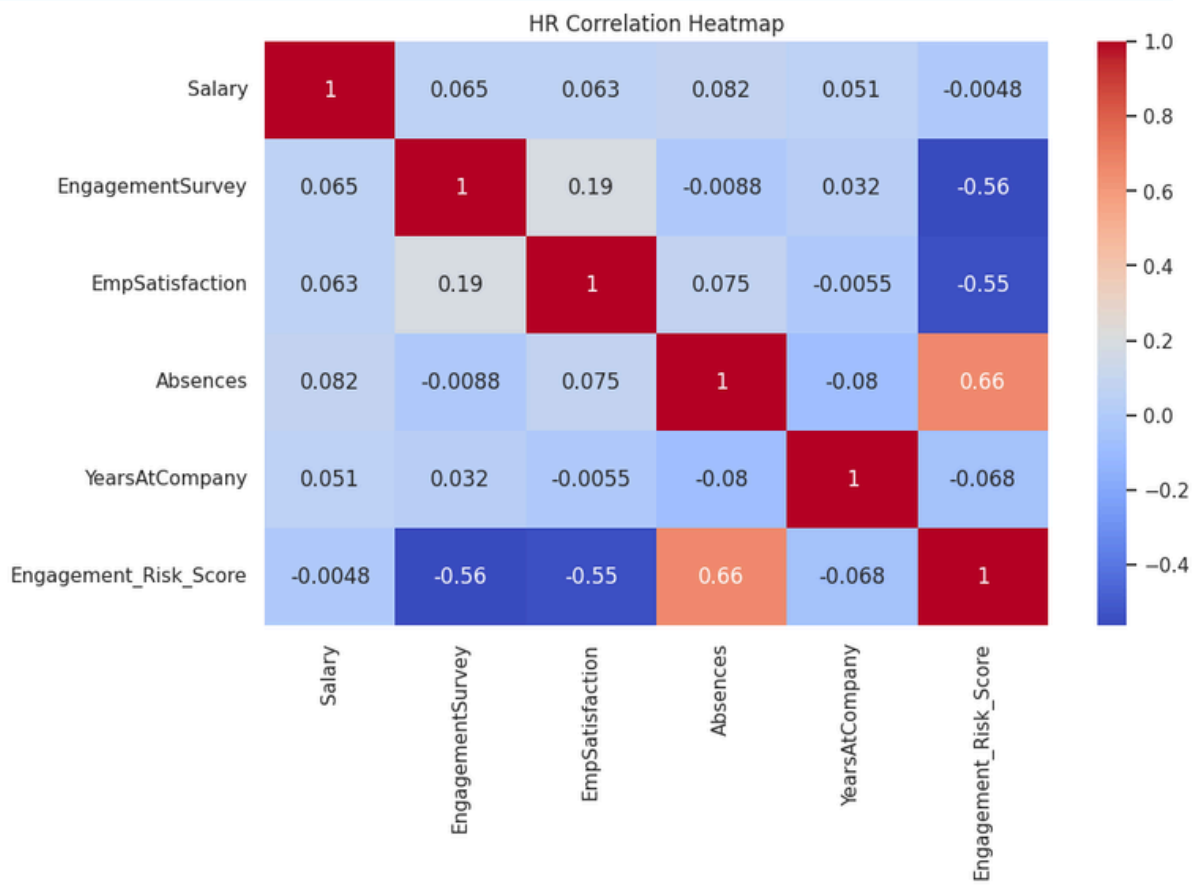
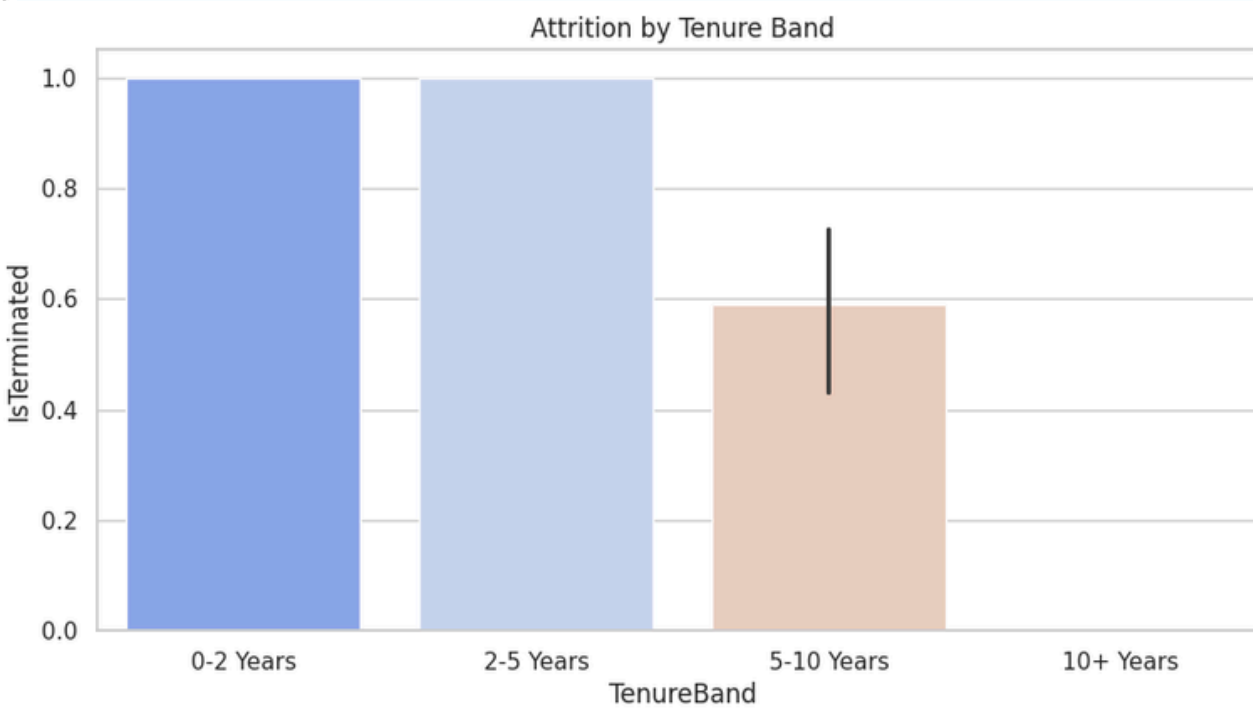
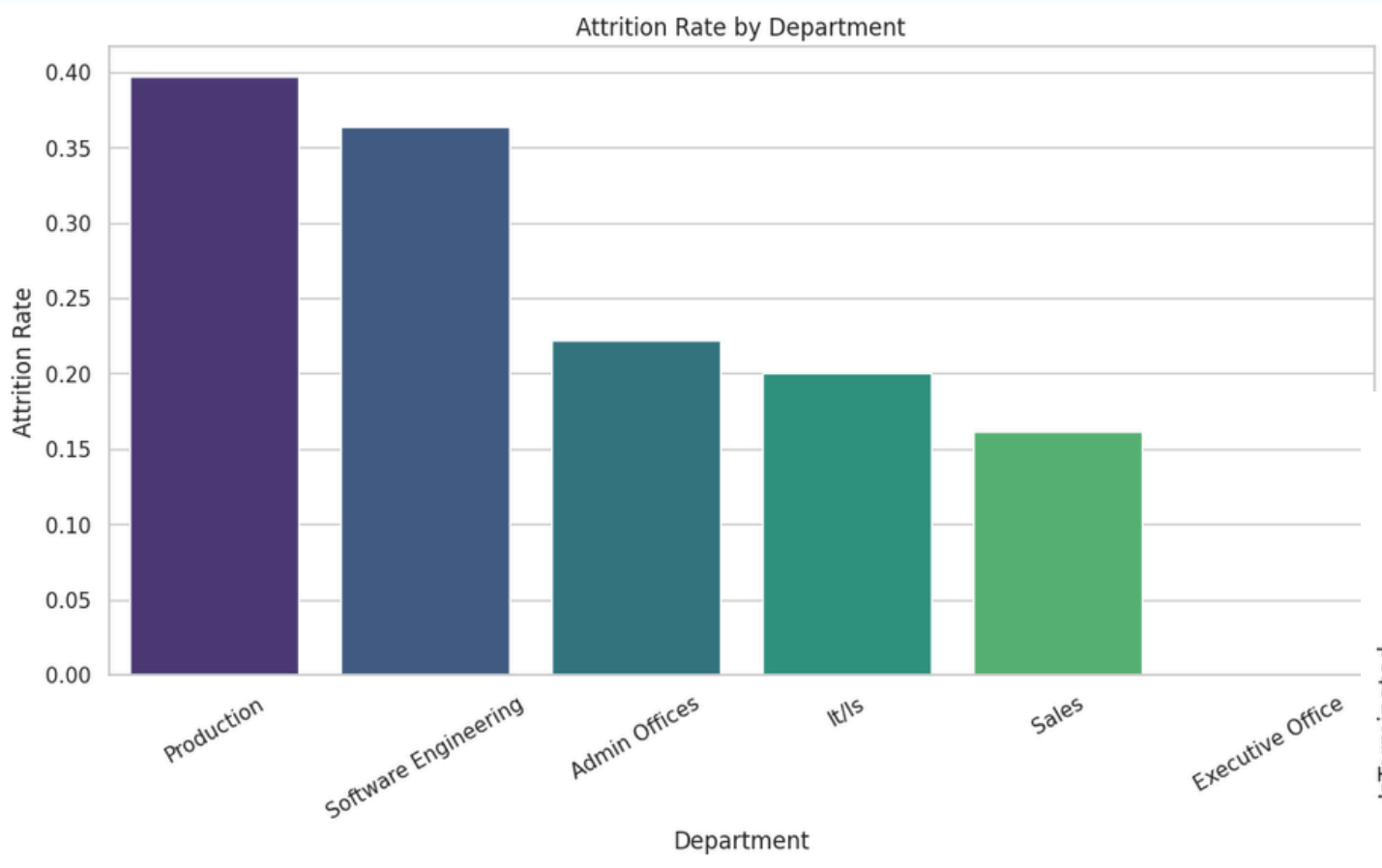
	LastPerformanceReview_Date	DaysLateLast30	Absences
0	1/17/2019	0	1
1	2/24/2016	0	17
2	5/15/2012	0	3
3	1/3/2019	0	15
4	2/1/2016	0	2

[5 rows x 36 columns]

WORKFORCE ANALYTICS

Key Workforce Insights

- Some departments showed significantly higher attrition
- Early-career employees had the highest resignation risk
- Engagement strongly influenced retention
- Compensation competitiveness affected turnover



SMART HIRING MODEL

Smart Hiring Machine

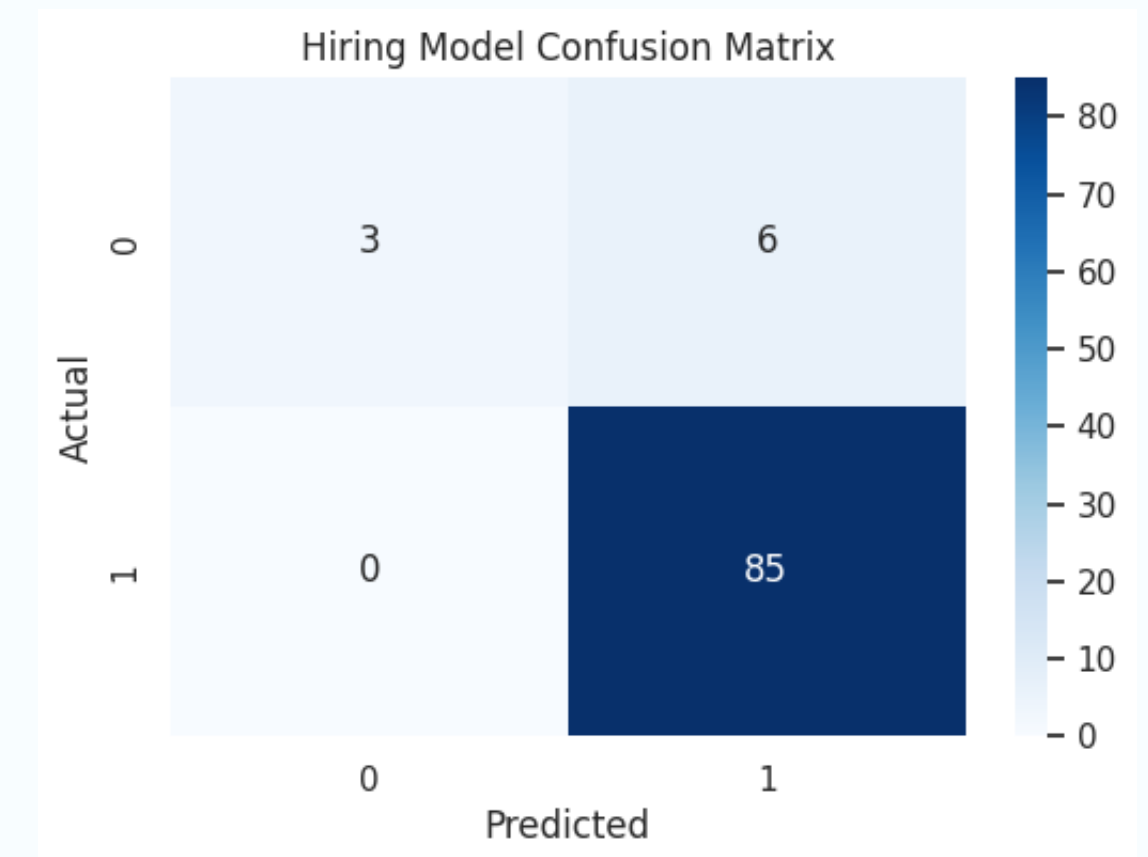
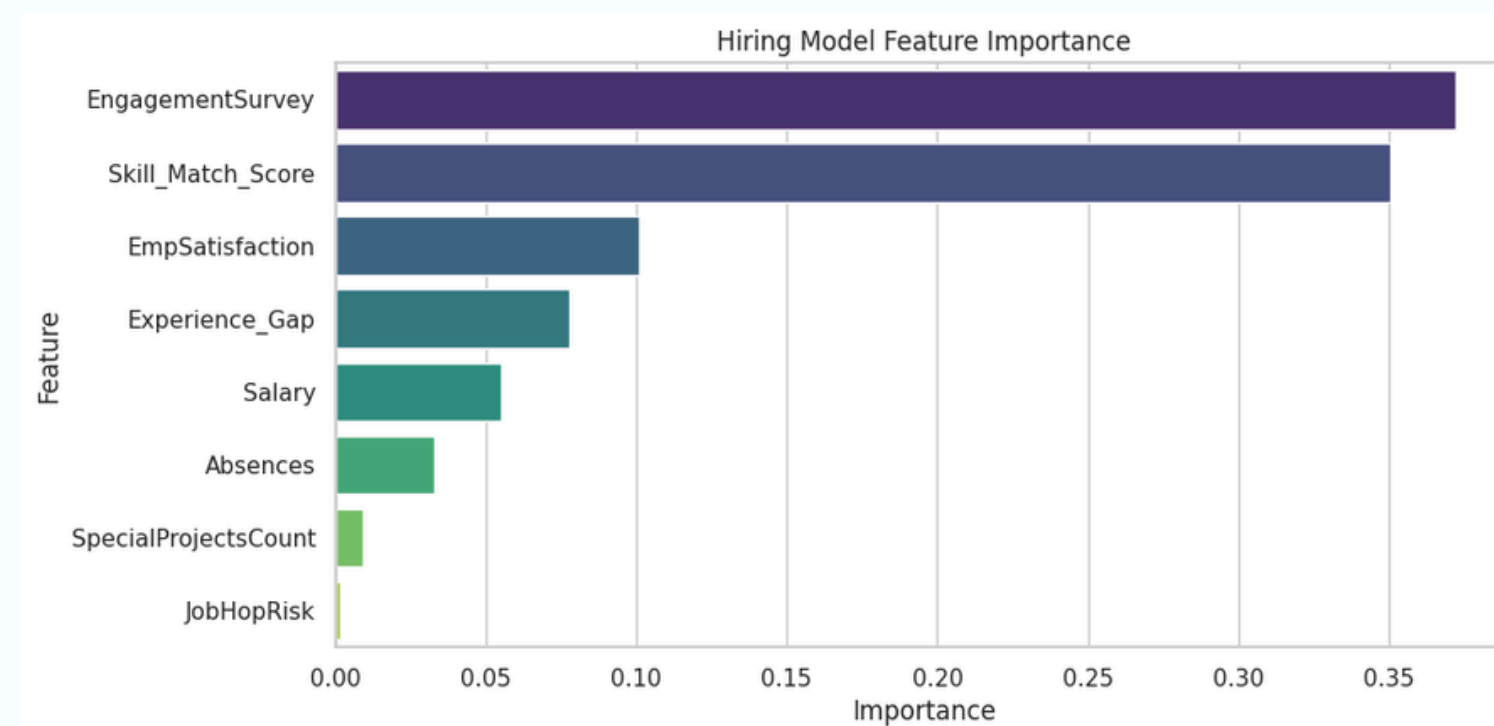
- Model Purpose:
- Predict high-potential candidates.

Main Factors

- Engagement
- Attendance
- Project participation
- Experience indicators

Outputs

- Hiring probability
- Candidate ranking
- Fairness auditing



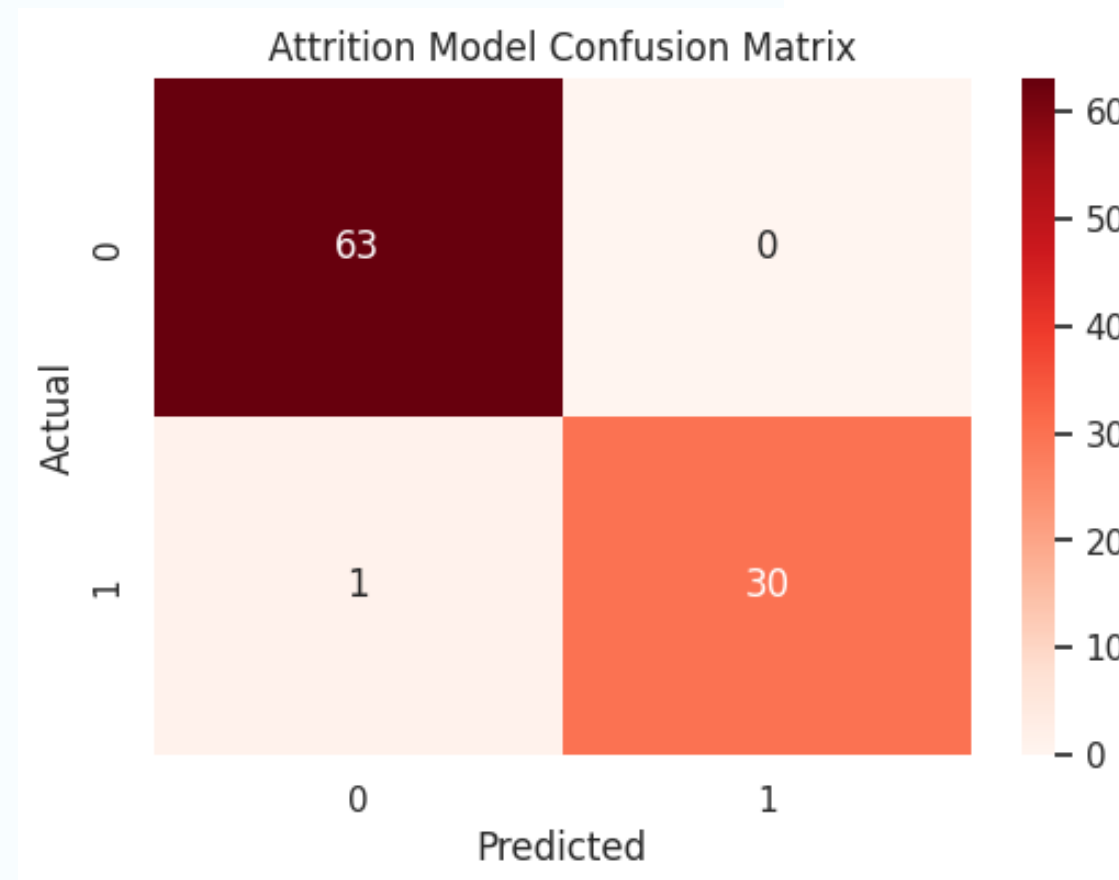
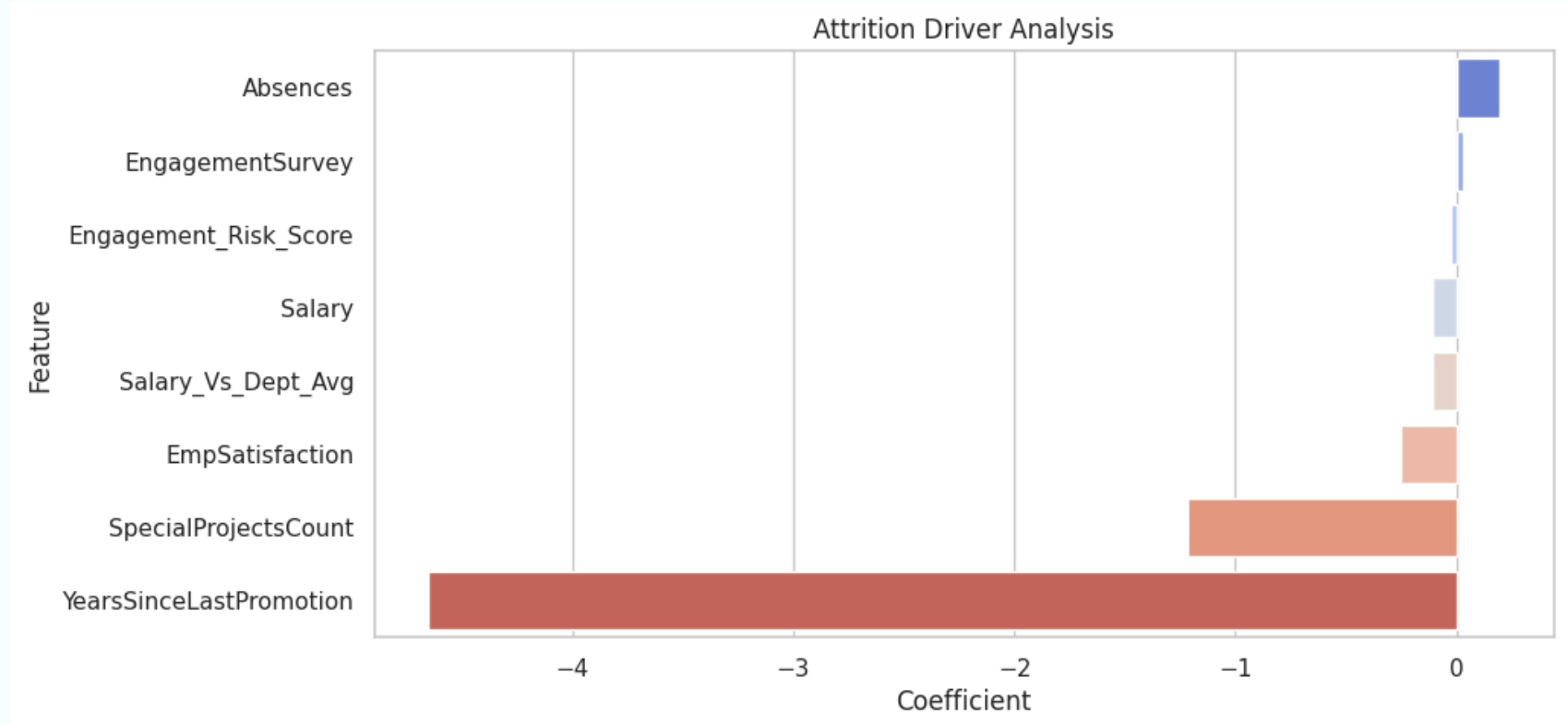
ATTRITION PREDICTION

Main Risk Drivers:

- Low engagement
- Low satisfaction
- High absences
- Weak salary competitiveness

Business Value

- Early intervention
- Reduced turnover
- Better retention planning



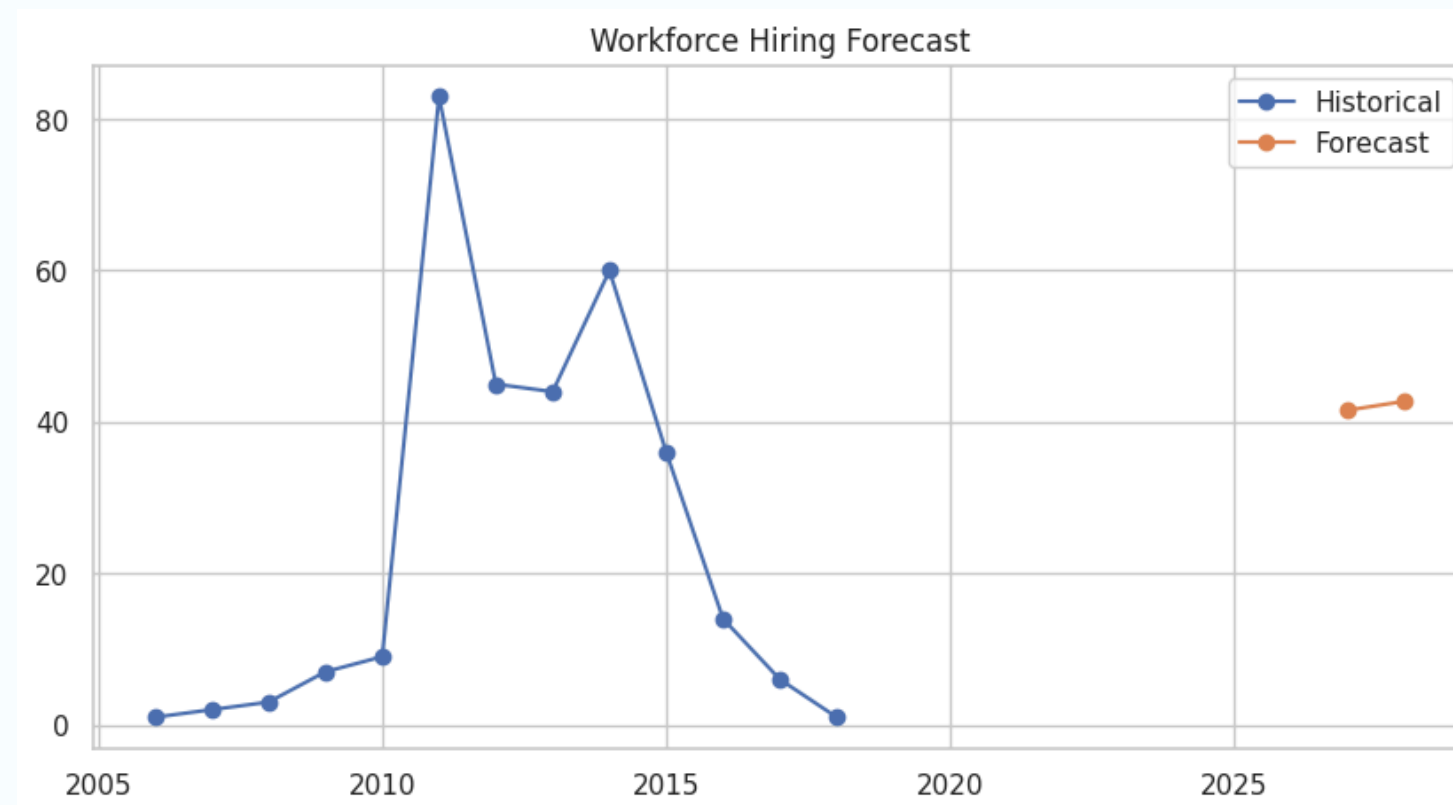
WORKFORCE SEGMENTATION

Clusters:

- Rising Stars
- Solid Contributors
- Needs Support

Forecasting

- Predict future hiring demand.



ETHICS & FAIRNESS

Responsible AI Governance

- ✓ Gender fairness audit
- ✓ Age fairness audit
- ✓ Explainable analytics
- ✓ GDPR-aligned data usage

Ethical Principles

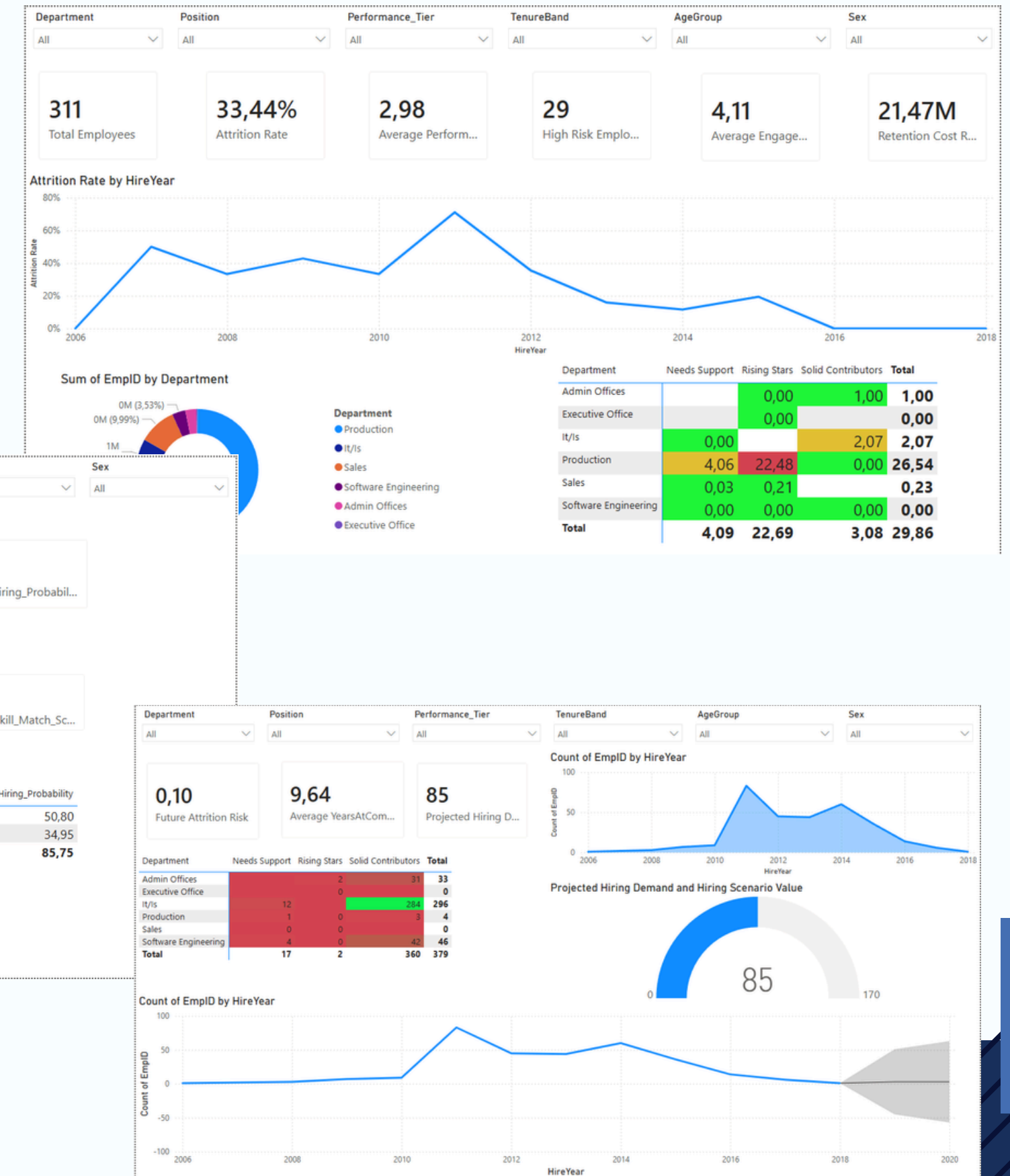
- Transparency
- Fairness
- Human oversight



POWER BI DASHBOARD

Dashboard Pages

1. Executive HR Command Center
2. Recruitment Intelligence
3. Attrition Risk Center
4. Performance Intelligence
5. Engagement & Wellbeing Monitor
6. Workforce Planning & Forecasting
7. Ethics & Fairness
8. HR Transformation Roadmap



RECOMMENDATIONS

- Improve retention initiatives for employees identified as high attrition risks through career development planning and manager support programs.
- Review compensation alignment for high-performing employees to reduce future turnover risk.
- Reduce overtime pressure and strengthen employee wellbeing initiatives in departments with lower satisfaction trends.
- Expand internal training and upskilling programs to address future workforce capability gaps.
- Continue regular fairness audits and AI governance reviews to maintain ethical and transparent HR decision-making.

The image features a light blue background with four decorative corner elements. Each corner contains a dark blue square with diagonal lines and a medium blue square. The top-left and bottom-right corners have the dark blue square in the foreground, while the top-right and bottom-left corners have the medium blue square in the foreground.

THANK YOU